

Midterm 1 will cover topics from 4.10 (in Vol. 1), 1.1, 1.2, 1.3, 1.5, and 3.1

- Problem 1 and Problem 2 - Riemann sum computation with specified  $n$ .
- Problem 3 and 4 - limits of Riemann sum set-up without evaluating.
- Problem 5 and Problem 6 - evaluate limit of Riemann sums and recognize them as definite integrals.
- Problem 7 - evaluate definite integral using the geometric meaning (i.e. net area) of definite integrals.
- Problem 8 - find derivatives by applying fundamental theorem of calculus Part 1.
- Problem 9 - indefinite and definite integrals.

1. Consider the function  $f(x) = x^2 - 2$  on the interval  $[2, 8]$ .

- 1). Sketch the graph of the function on the given interval.
- 2). Illustrate the left Riemann sum on the given interval with  $n = 3$  by sketching the appropriate rectangles on the graph of the function.
- 3). Calculate the left Riemann sum on the given interval with  $n = 3$ .

2. Consider the function  $f(x) = -x^2 + 3$  on the interval on  $[-1, 3]$ .

- 1). Sketch the graph of the function on the given interval.
- 2). Illustrate the right Riemann sum on the given interval with  $n = 4$  by sketching the appropriate rectangles on the graph of the function.
- 3). Calculate the right Riemann sum on the given interval with  $n = 4$ .

3. Consider the function  $f(x) = 3 + \ln x$  on  $[0, 4]$ .

- 1). If the interval  $[0, 4]$  is divided into  $n$  subintervals of equal length, what is the width of the each subinterval?
- 2). Find the right endpoint  $x_i$  of the  $i$ th subinterval.
- 3). Express  $\int_0^4 (3 + \ln x) dx$  as a limit of Riemann sums with right endpoints. *Note: You do not need to simplify and evaluate the limit.*

4. Consider the function  $f(x) = x\sqrt{x+2}$  on  $[1, 3]$ .

- 1). If the interval  $[1, 3]$  is divided into  $n$  subintervals of equal length, what is the width of the each subinterval?
- 2). Find the right endpoint  $x_i$  of the  $i$ th subinterval.
- 3). Express  $\int_1^3 x\sqrt{x+2} dx$  as a limit of Riemann sums with right endpoints. *Note: You do not need to simplify and evaluate the limit.*

5. 1). Simplify the sum  $R_n = \sum_{i=1}^n \frac{2}{n} \left( 2 - \frac{2i}{n} + \frac{8i^3}{n^3} \right)$ . Write your answer as a function of  $n$  without any summation signs.

2). Use your result from Part 1), evaluate  $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{2}{n} \left( 2 - \frac{2i}{n} + \frac{8i^3}{n^3} \right)$ .

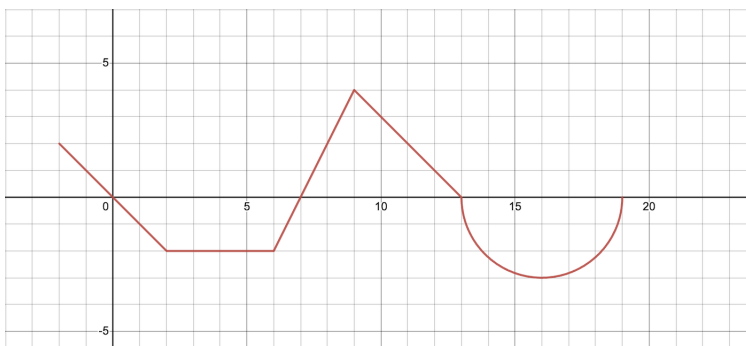
3). Express  $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{2}{n} \left( 2 - \frac{2i}{n} + \frac{8i^3}{n^3} \right)$  as a definite integral:  $\int_{\underline{\quad}}^{\underline{\quad}} \underline{\hspace{2cm}} dx$ .

6. 1). Simplify the sum  $R_n = \sum_{i=1}^n \frac{1}{n} \left( \frac{2i}{n} + \frac{i^2}{n^2} \right)$ . Write your answer as a function of  $n$  without any summation signs.

2). Use your result from Part 1), evaluate  $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{n} \left( \frac{2i}{n} + \frac{i^2}{n^2} \right)$ .

3). Express  $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{n} \left( \frac{2i}{n} + \frac{i^2}{n^2} \right)$  as a definite integral:  $\int_{\underline{\quad}}^{\underline{\quad}} \underline{\hspace{2cm}} dx$ .

7. Consider the  $f(x)$ , whose graph is given as follows. Evaluate the following integrals.



1)  $\int_{-2}^2 f(x) dx$

2)  $\int_0^7 f(x) dx$

3)  $\int_6^9 f(x) dx$

4).  $\int_7^{16} f(x) dx$

5).  $\int_9^{19} f(x) dx$

6).  $\int_{-2}^{19} f(x) dx$

8. Find the derivative  $F'(x)$  of each of the following functions.

1).  $F(x) = \int_1^x t^3 dt$

2).  $F(x) = \int_5^{\sqrt{x}} \frac{\cos t^2}{t^4} dt$

3).  $F(x) = \int_{3x}^{-5} \sin(t^2) dt$

4).  $F(x) = \int_{\ln x}^{e^x} 7 \sin t dt$

9. Find the following integrals.

1).  $\int (-4 \sin x + 6 \cos x) dx$

3).  $\int_{-e}^{-1} \frac{10}{x} dx$

5).  $\int \frac{x^5 \sqrt{x} + 1}{x^4} dx$

7).  $\int_1^6 \frac{3}{2x-1} dx$

9).  $\int_0^{\pi/6} \sec^2(2x) dx$

11).  $\int \frac{\cos x}{1 + \sin^2 x} dx$

13).  $\int \frac{x}{(1 - 2x^2)^5} dx$

15).  $\int_1^e \frac{(\ln x)^2}{x} dx$

17).  $\int_0^{\pi/2} \frac{\sin x}{1 + \cos x} dx$

19).  $\int_0^{\pi/3} \cos^4 \theta \sin \theta d\theta$

21).  $\int x e^{-2x} dx$

22).  $\int x \cos(4x) dx$

23).  $\int x^2 \ln x dx$

2).  $\int \frac{5 - 2xe^x}{x} dx$

4).  $\int e^{-3x+2} dx$

6).  $\int_1^4 \frac{2x^2 + 2x - \sqrt{x}}{\sqrt{x}} dx$

8).  $\int_1^{\sqrt{3}} \frac{3}{1 + x^2} dx$

10).  $\int_0^{\pi/6} \sec(2x) \tan(2x) dx$

12).  $\int x^2 e^{-4x^3} dx$

14).  $\int \frac{(\sin^{-1} x)^8}{\sqrt{1 - x^2}} dx$

16).  $\int_1^2 \frac{e^{1/x}}{x^2} dx$

18).  $\int \frac{(\sqrt{x} + 3)^3}{\sqrt{x}} dx$

20).  $\int_0^1 \frac{(\tan^{-1} x)^3}{1 + x^2} dx$

22).  $\int x \sin(2x) dx$

23).  $\int \sin^{-1} x dx$

24).  $\int \frac{\ln x}{\sqrt{x}} dx$

## Formula Sheet

$$\sum_{i=1}^n c = cn$$

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6} = \frac{2n^3 + 3n^2 + n}{6}$$

$$\sum_{i=1}^n i^3 = \left[ \frac{n(n+1)}{2} \right]^2 = \frac{n^4 + 2n^3 + n^2}{4}$$

$$\int \cot x \csc x \, dx = -\csc x + C, \quad \int \csc^2 x \, dx = -\cot x + C$$

$$\int \tan x \, dx = \ln |\sec x| + C, \quad \int \cot x \, dx = \ln |\sin x| + C$$

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C, \quad \int \csc x \, dx = \ln |\csc x - \cot x| + C$$

$$\int \frac{1}{1+x^2} \, dx = \arctan x + C, \quad \int \frac{1}{\sqrt{1-x^2}} \, dx = \arcsin x + C$$

$$\int \frac{1}{a^2+x^2} = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C, \quad \int \frac{1}{\sqrt{a^2-x^2}} = \arcsin\left(\frac{x}{a}\right) + C$$

| $\theta$      | 0 | $\frac{\pi}{6}$      | $\frac{\pi}{4}$      | $\frac{\pi}{3}$      | $\frac{\pi}{2}$ |
|---------------|---|----------------------|----------------------|----------------------|-----------------|
| $\sin \theta$ | 0 | $\frac{1}{2}$        | $\frac{\sqrt{2}}{2}$ | $\frac{\sqrt{3}}{2}$ | 1               |
| $\cos \theta$ | 1 | $\frac{\sqrt{3}}{2}$ | $\frac{\sqrt{2}}{2}$ | $\frac{1}{2}$        | 0               |
| $\tan \theta$ | 0 | $\frac{\sqrt{3}}{3}$ | 1                    | $\sqrt{3}$           | undefined       |